Distractor interference in Sirius

Comparing psychometric models with full-length chains, simpler models and extensions

Analysis of Vera Schroeder's data and models, prepared for Krešimir Josić · 3 November 2026

Summary

All 28 models were fit with Vera's chain length (4 chains × 5000 draws) and scored two ways: trial-level LOO, as in her reports, and leave-one-session-out (LOSO), which asks how well a model predicts a session it has not seen. New since the last report: simpler versions of Model D, extensions in which the monkey's criterion changes over the study, a shared daily bias and a second capture model.

1. **Longer chains change nothing.** Scores moved by at most 1.0 elpd units (trial level) and 0.7 (new sessions); the only change of order is between two models that were within half a unit of each other.
2. **Model D still beats Vera's models**: by 12.7 ± 8.2 (trial level) and 19.6 ± 9.0 (new sessions) over B, and by about 50 over C. The gain comes from bimanual lapse rates that depend on the distractor.
3. **Sirius's criterion moved over the study.** Letting the session bias follow the distractor amplitude, or drift over time, is the largest improvement found (16.2 ± 4.6 over D for new sessions). The two versions cannot be told apart, because the amplitude was raised steadily during the study. The unimanual PSE rose by about 3.0 µm (left hand) and 6.1 µm (right).
4. **Fewer parameters work.** The distractor terms, and largely the slope, can be the same in every session; the bias and the lapse rates must vary. **Model E** (D with fixed distractor terms, plus the trend; 12 instead of 16 parameters per session) predicts as well as D+crit with 20 fewer effective parameters, and no model predicts clearly better. With only one bias per hand and session, plus the trend, a model still predicts new sessions better than B.
5. **The distractor makes the target feel stronger**: on bimanual trials the PSE shifts by −4.0 µm (left hand) and −5.9 µm (right) on average, more for stronger distractors. For the left hand the shift is proportional to the distractor amplitude; the right hand has an extra shift whatever the amplitude. There is no change in sensitivity.
6. **The extra bimanual errors look like answers driven by the distractor**: for the left hand they occur only with strong distractors, and they are nearly as fast as correct answers.

Recommendation: use Model E, report both scores, and treat "adaptation to the distractor" versus "slow drift" as unresolved by this design (Section 5).

1 Data, models and how they were compared

Data. The analysis uses the 12,937 trials of monkey Sirius that Vera selected: 42 sessions from 30 March to 31 July 2023, both hands tested in every session, about 80 trials per hand and condition in each session. A vibration of 6–50 µm is applied to one hand and the monkey reports whether it is stronger or weaker than the trained boundary of 28 µm. On half of the trials (bimanual) the other hand receives a distractor vibration at the same time. The distractor amplitude is fixed within a session and was raised over the study, from 6 µm in the first eight sessions to 32 µm in most of the later ones (Fig. 4, top).

Models. All models share one psychometric curve with "high" and "low" lapse rates (Appendix A). They differ in what is allowed to change between hands, conditions and sessions. Vera's Model B fits unimanual and bimanual trials of each hand as unrelated data sets. Her Model C fits one curve

per hand and shifts it on bimanual trials by an amount proportional to the distractor amplitude, with the same lapse rates in both conditions. Model D, recommended in the previous report, is C with three additions: bimanual trials get their own lapse rates (L), these change with the distractor amplitude (K), and a distractor of any amplitude can shift the curve (presence, P). This report adds simpler versions of D, which keep parameters fixed across sessions, and extensions of D that ask whether the monkey's criterion moved between sessions in a systematic way. The recommended Model E combines one of each. Table 1 lists all 28 models.

Sampling. Every model was refit with Vera's setting: 4 chains of 5000 draws after 1000 tuning steps, target acceptance 0.9. Model C and its variants were sampled with nutpie, a faster implementation of the same NUTS algorithm; A and the B family with PyMC's own sampler, which handles B's parameterization better. The model code is the same in both cases, and every model's likelihood was checked against an independent implementation. All fits are clean (Appendix B): 0 divergences in total, $\hat{R} \leq 1.004$, and at least 1735 effective draws for every parameter.

Two ways to score a model. Both scores estimate the expected log predictive density (elpd) of data the model has not seen; higher is better, and differences are in log-likelihood units summed over trials.

- **Trial-level LOO** is the criterion in Vera's reports (PSIS-LOO). It predicts each trial from all the others, including the other trials of the same session, so it asks how well the model describes the sessions it was fit to. Trials from one session share parameters, so the usual standard error, which treats trials as independent, is unreliable (for the larger differences here it is too small); the standard error of every difference is computed from the 42 per-session sums instead.
- **Leave-one-session-out (LOSO)** predicts every trial of a session from the other 41 sessions, with that session's own parameters drawn from the population distribution. It asks how well the model would predict a new session, which is what claims about the population-level effects (the shift per µm of distractor, the presence offset) rest on. It is computed without refitting, by importance sampling over the held-out session's parameters followed by PSIS over the posterior draws (1000 draws thinned from the 20,000). In checks with exact refits without the held-out session, the two agreed to within 0.2 elpd, including sessions with Pareto k up to 1.03 (Appendix B).

A difference is treated as clear when it exceeds about twice its standard error.

2 Results

2.1 Longer chains change nothing material

For the 14 models fit both ways, the trial-level elpd changed by at most 1.04 (median 0.37) and the LOSO elpd by at most 0.67 (median 0.30). No pair of models changed order under trial-level LOO, and one pair changed order under LOSO (C+L and C+LMmax, 0.4 apart before and 0.4 after). The conclusions of the previous report that do not involve the new models therefore stand. All numbers below are from the 5000-draw fits.

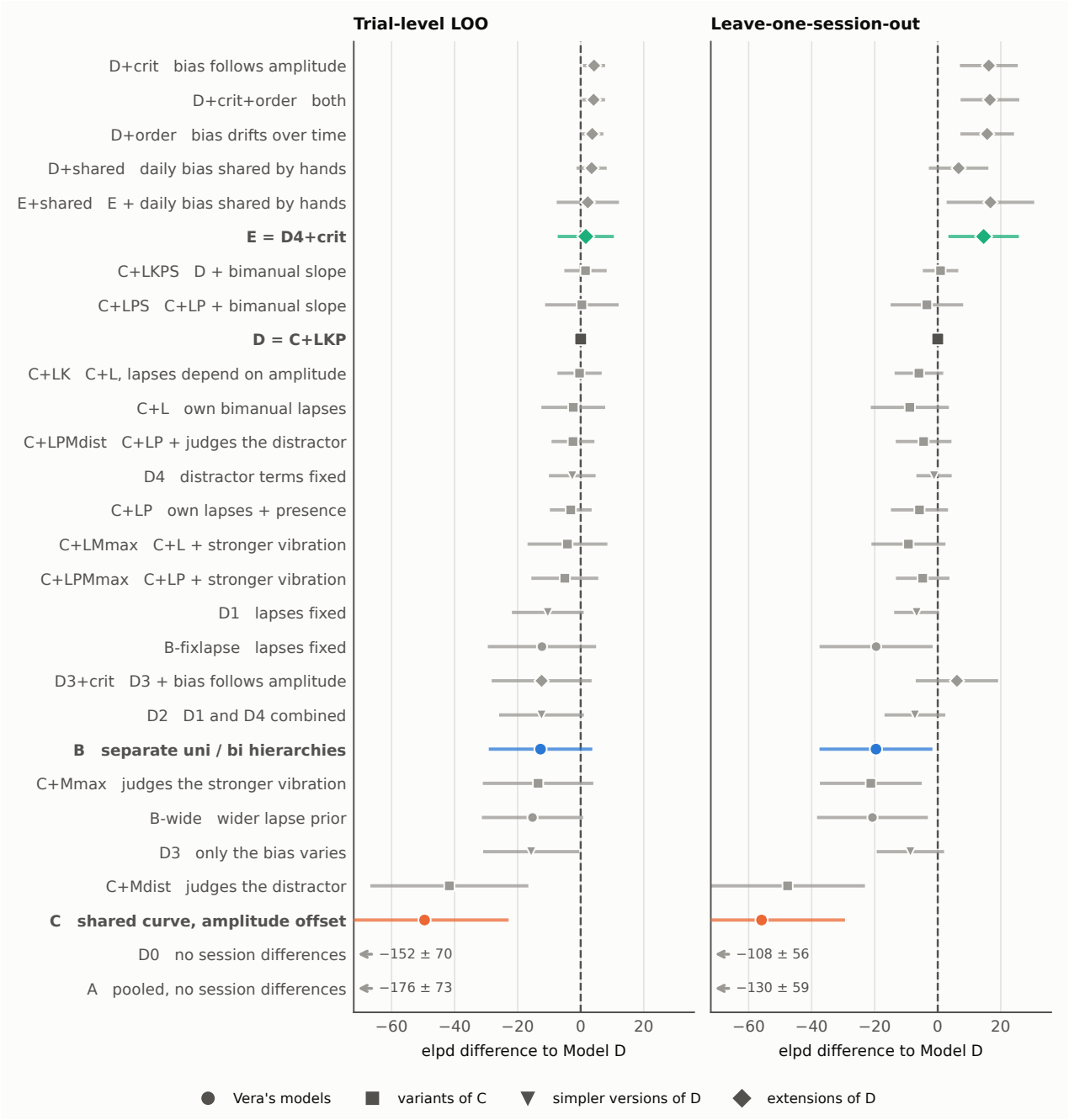


Figure 1. Predictive performance of every model relative to Model D (dashed line), with ±2 standard errors; right of the line is better than D. Left: trial-level LOO, the criterion in Vera's reports, with standard errors from per-session sums. Right: leave-one-session-out, the accuracy for a session the model has not seen. Models are sorted by the left panel. B, C and the recommended Model E are coloured; arrows mark models off the scale.

Table 1. All models, their size and their predictive performance relative to Model D. Parameters: hand- or group-level parameters / parameters per session. p_{loo}: effective number of parameters from trial-level LOO. Differences are in elpd (log-likelihood units summed over all trials); positive means better than D; standard errors in brackets.

Model	What it is	Parameters pop. / session	p_{loo}	Trial LOO vs D (SE)	LOSO vs D (SE)
Vera's models					
A	pooled: no session differences	16 / 0	13	−176.1 (36.5)	−130.0 (29.6)
B	separate unimanual and bimanual hierarchies	32 / 16	179	−12.7 (8.2)	−19.6 (9.0)
B-wide	B with a 3× wider prior on the lapse sd	32 / 16	196	−15.3 (8.0)	−20.7 (8.8)
B-fixlapse	B with lapses fixed per group	24 / 8	169	−12.3 (8.6)	−19.6 (9.0)
C	shared curve, bimanual offset ∝ amplitude	20 / 10	139	−49.5 (13.3)	−55.9 (13.2)
Variants of C					
C+L	C + own lapse rates on bimanual trials	28 / 14	142	−2.4 (5.1)	−8.9 (6.2)
C+LK	C+L + bimanual lapses depend on amplitude	32 / 14	143	−0.4 (3.5)	−6.0 (3.9)
C+LP	C+L + presence offset	32 / 16	147	−3.2 (3.3)	−5.8 (4.5)
C+LPS	C+LP + bimanual slope change	36 / 18	155	0.4 (5.8)	−3.4 (5.8)
C+LKPS	D + bimanual slope change	40 / 18	156	1.5 (3.4)	0.9 (2.8)
C+Mmax	C + judges the stronger vibration	22 / 10	134	−13.5 (8.8)	−21.2 (8.1)
C+Mdist	C + judges the distractor	22 / 10	135	−41.7 (12.5)	−47.6 (12.3)
C+LMmax	C+L + judges the stronger vibration	30 / 14	138	−4.2 (6.3)	−9.3 (5.9)
C+LPMmax	C+LP + judges the stronger vibration	34 / 16	141	−5.0 (5.3)	−4.8 (4.2)
C+LPMdist	C+LP + judges the distractor	34 / 16	141	−2.4 (3.4)	−4.5 (4.4)
Model D, recommended in the previous report					
D (= C+LKP)	C+LK + presence offset	36 / 16	145	0	0
Simpler versions of D					
D4	D, distractor terms fixed per hand	32 / 12	125	−2.7 (3.7)	−1.2 (2.8)
D1	D, lapses fixed per hand and condition	28 / 8	130	−10.4 (5.7)	−6.7 (3.6)
D2	D1 and D4 combined	24 / 4	109	−12.4 (6.7)	−7.2 (4.8)
D3	D2 + slope fixed: only the bias varies	22 / 2	88	−15.7 (7.6)	−8.7 (5.3)
D0	D with no session differences	20 / 0	18	−152.0 (35.2)	−108.2 (28.2)
Extensions of D					
D+shared	D + daily bias shared by the two hands	37 / 17	141	3.5 (2.4)	6.6 (4.7)
D+crit	D + bias follows the distractor amplitude	38 / 16	144	4.2 (1.8)	16.2 (4.6)
D+order	D + bias drifts over the study	38 / 16	144	3.7 (1.8)	15.7 (4.3)
D+crit+order	D + both	40 / 16	144	4.1 (1.8)	16.6 (4.7)
Simpler versions with the trend					
E (= D4+crit)	D4 + bias follows the distractor amplitude	34 / 12	123	1.6 (4.5)	14.6 (5.6)
D3+crit	D3 + bias follows the distractor amplitude	24 / 2	84	−12.4 (7.9)	6.1 (6.5)
E+shared	E + daily bias shared by the two hands	35 / 13	121	2.3 (4.9)	16.7 (6.9)

2.2 Model D beats Vera's models, and why

Model D predicts better than Model B by both scores (Fig. 1, Table 1): by 12.7 (SE 8.2) at the trial level and by 19.6 (SE 9.0) for new sessions, where it predicts 29 of the 42 held-out sessions better. It does this with fewer effective parameters (p_{loo} 145 against 179), because it ties the unimanual and bimanual trials of a hand to one curve instead of fitting them separately. Model C is far behind both (D minus C: 49.5 $\pm$ 13.3 at the trial level, 55.9 $\pm$ 13.2 for new sessions).

The variants of C show which parts of D matter.

- Giving bimanual trials their own lapse rates (C+L) is the largest single step: 47.2 over C at the trial level. C+L alone already beats B (10.4 $\pm$ 7.7; new sessions 10.7 $\pm$ 8.4).
- Letting those lapse rates change with the distractor amplitude and adding the presence offset helps most for new sessions: D minus C+L is 2.4 $\pm$ 5.1 at the trial level and 8.9 $\pm$ 6.2 for new sessions.
- A change of slope on bimanual trials adds nothing (D with a bimanual slope, minus D: 1.5 at the trial level, 0.9 for new sessions).
- Capture by the stronger vibration explains much of C's deficit with one parameter per hand (36.0 over C), but adds nothing once the lapses are per condition (-1.8 ± 5.0 over C+L at the trial level). Replacing D's amplitude-dependent lapses by capture of the distractor (C+LPMdist: on a fraction π of bimanual trials Sirius judges the distractor) does slightly worse than D (-2.4 ± 3.4 at the trial level, -4.5 ± 4.4 for new sessions), and puts capture on only 1.5% (left) and 1.0% (right) of bimanual trials.
- Widening the prior on B's lapse variation does not help B (-2.5 at the trial level, -1.1 for new sessions), and fixing B's lapses across sessions makes no difference (0.5 at the trial level, 0.0 for new sessions).

The raw counts show what B and C miss (Fig. 2). For weak targets (6–12 µm) Sirius answered "high" on 81 of 1,606 bimanual trials (5.0%) but on only 4 of 1,627 unimanual trials (0.25%). Strong targets (44–50 µm) drew "low" on 2.0% of trials in both conditions. The bimanual excess differs between the hands. For the left hand it appears only with strong distractors: none in 262 trials of the 6 and 12 µm sessions, 36 of 447 (8.1%) in the 24–32 µm sessions. For the right hand it is about 5% at every distractor amplitude. Model C has one pair of lapse rates for both conditions, so it cannot produce twenty times more weak-target errors on bimanual trials. Model B gives bimanual trials their own lapse rates but cannot tie them to the distractor: it pulls every session toward a common rate, so for the left hand it predicts 3.0% "high" answers in the 6–12 µm sessions, where there were none. In D the bimanual lapse rate changes with the session's distractor amplitude. For the left hand the "high" lapse rises steeply with amplitude ($\kappa_h = 2.42$, 95% interval 1.09 to 3.87, on the logit scale per 14.6 µm); for the right hand it barely changes ($\kappa_h = 0.56$, -0.37 to 1.49). In posterior predictive checks of these error rates by hand and amplitude the observed rate falls inside the 95% predictive interval in 20 of 20 cells for D, 20 for E, 19 for B and 17 for C. Figure 3 shows where the gains over B come from: the sessions with the weakest distractors and the 24–30 µm sessions. In the 32 µm sessions B still fits slightly better at the trial level, but not for new sessions.

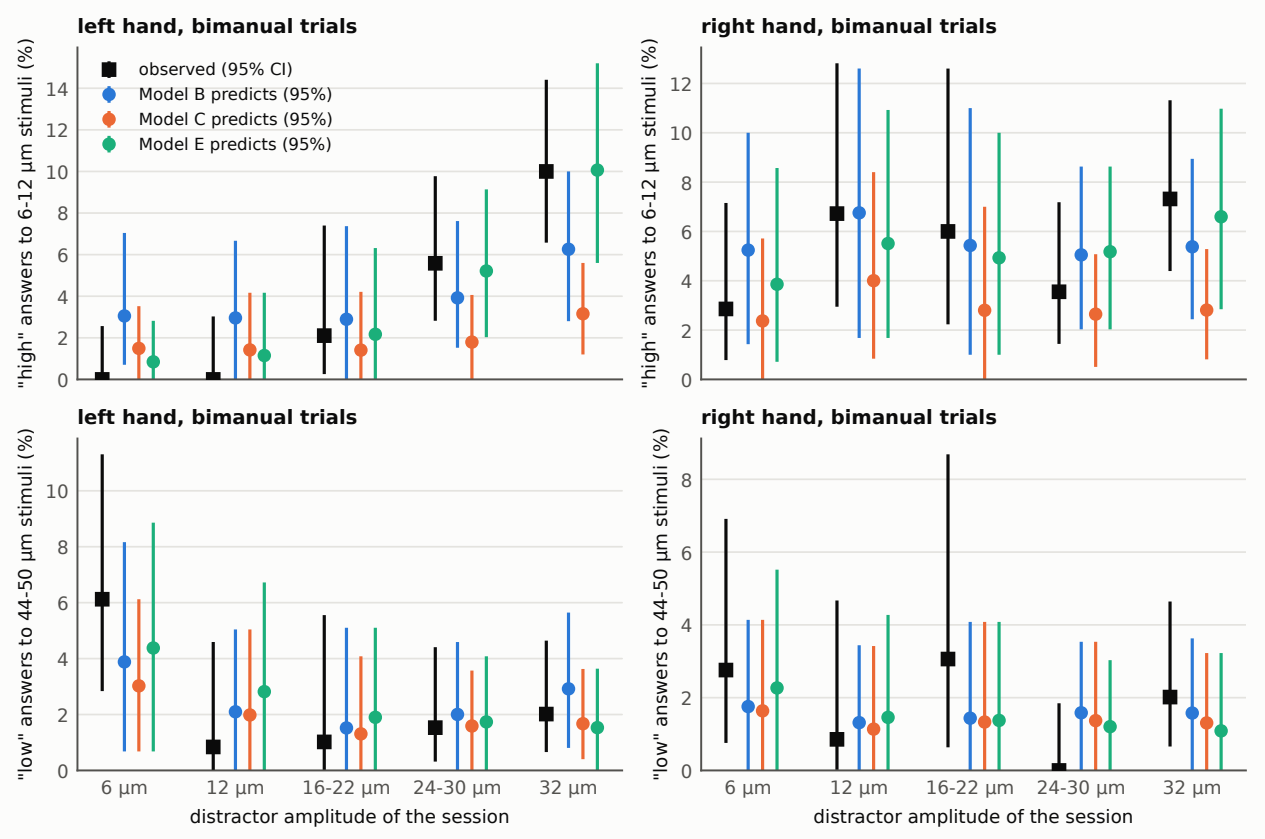


Figure 2. Lapse-like answers on bimanual trials, by hand and by the session's distractor amplitude: "high" answers to the two weakest targets (top) and "low" answers to the two strongest (bottom). Black: observed rate with a 95% binomial interval. Coloured: each model's posterior predictive mean and 95% interval.

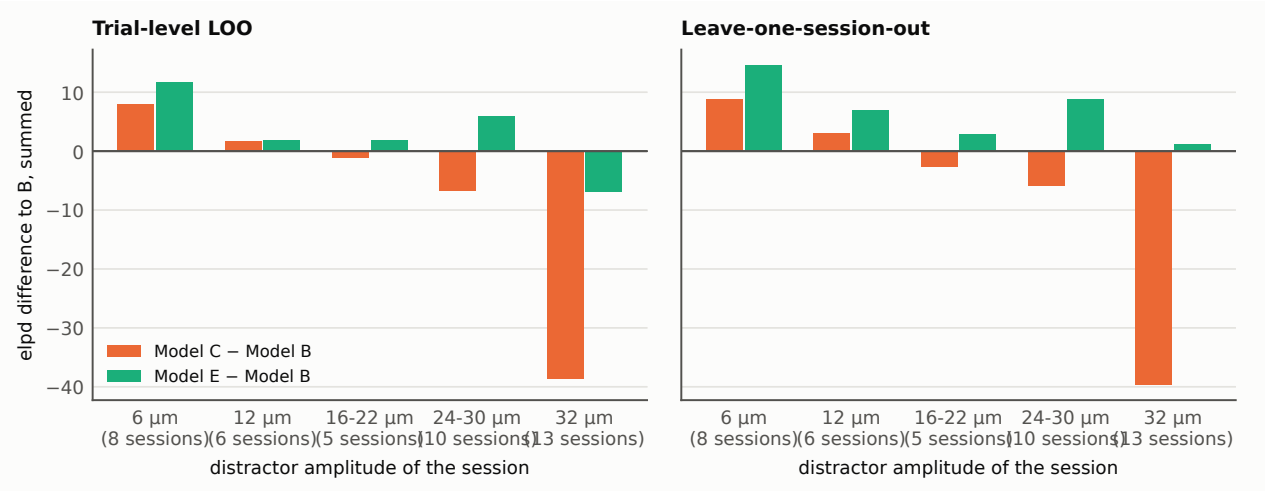


Figure 3. Where Models C and E gain or lose against Model B, summed over the sessions with each distractor amplitude. Positive: better than B.

2.3 Extensions: Sirius's criterion moved over the study

Model D gives every session its own bias, but it treats the sessions as interchangeable: it expects no order in them. There is order (Fig. 4). Sirius's unimanual PSE rose over the study, most clearly for the right hand: it called fewer unimanual targets "high" in the later sessions, which were also the sessions with the strongest distractors.

Two extensions describe this with one extra parameter per hand. In **D+crit** the session bias follows the session's distractor amplitude; in **D+order** it drifts linearly with the session's place in the study.

- Both predict clearly better than D, above all for new sessions: D+crit by 4.2 $\pm$ 1.8 at the trial level and 16.2 $\pm$ 4.6 for new sessions (better on 34 of 42), D+order by 3.7 $\pm$ 1.8 and 15.7 $\pm$ 4.3. No other single addition to D comes close; the next best, a shared daily bias, gains 6.6 for new sessions.
- The two cannot be told apart. D+crit minus D+order is 0.6 $\pm$ 1.1 at the trial level and 0.5 $\pm$ 3.8 for new sessions. A model with both terms is no better than either (0.4 $\pm$ 2.1 over D+crit), and in it neither coefficient is clearly different from zero. The design does not allow the distinction: the distractor amplitude was raised steadily, so it is correlated 0.91 with the session order. Only the four 24 µm sessions run late in the study break the pattern, and they do not settle it: three lean slightly towards drift and one towards adaptation (together -1.5 elpd for D+crit against D+order).
- The size of the change: in a typical session the unimanual PSE is 3.0 µm higher in the 32 µm sessions than in the 6 µm sessions for the left hand (95% interval 1.0 to 4.9) and 6.1 µm higher for the right (3.9 to 8.2), comparable to the bimanual shift itself. For the left hand the rise roughly offsets the distractor's push: the overall proportion of "high" answers to left-hand targets stays between 0.44 and 0.49 in every amplitude group, while unimanual "high" answers fall from 0.44 to 0.39 and bimanual ones rise from 0.45 to 0.54. For the right hand the criterion rose further than that.

Modelling the trend changes one conclusion of the previous report. When every session's bias is pulled toward a single value (Model D), part of the trend is absorbed by the bimanual terms, because the bias is shared by a session's unimanual and bimanual trials. With the trend modelled, the bimanual shift grows more steeply with amplitude ($b_2 = 0.72$ left and 0.68 right, against 0.54 and 0.49 in D) and the presence offset shrinks. For the right hand it remains clear ($b_3 = 0.75$, 0.18 to 1.33); for the left hand it is no longer distinguishable from zero (0.16, -0.44 to 0.78). The shift averaged over sessions, the JND result and the lapse pattern do not change (Table 2).

A shared daily bias. The two hands' session biases are correlated (0.50 between their unimanual PSEs in Model D, 0.36 to 0.41 after a linear trend in amplitude or time is removed). A daily offset shared by the hands (D+shared) improves on D (3.5 $\pm$ 2.4 at the trial level, 6.6 $\pm$ 4.7 for new sessions), but much less than the trend does for new sessions (D+shared minus D+crit: -9.6 ± 4.1): much of what the hands share from day to day is the trend. Added to E, the shared offset is small ($\sigma_v = 0.47$ on the logit scale) and gains little (0.6 $\pm$ 1.8 at the trial level, 2.2 $\pm$ 2.8 for new sessions).

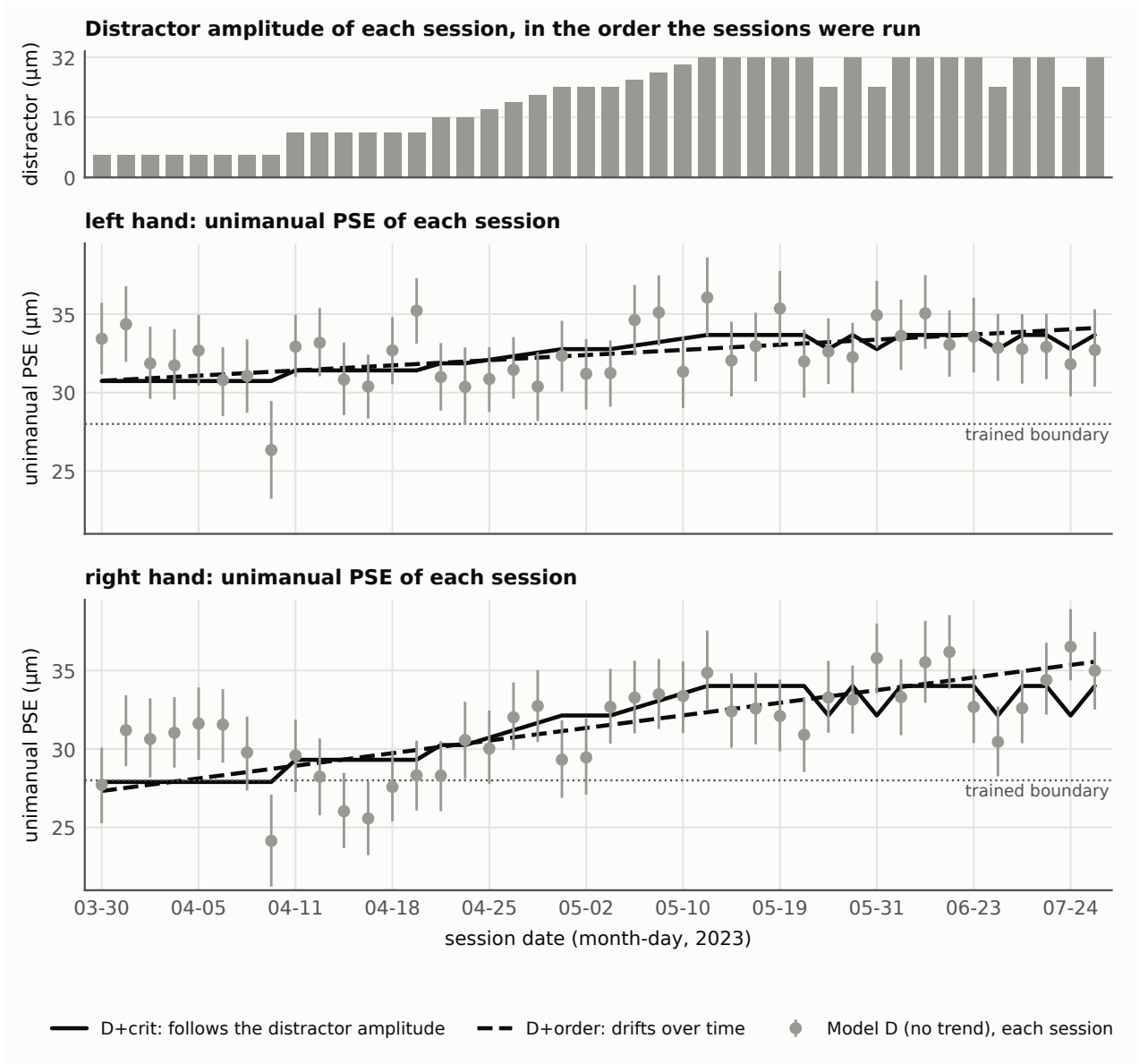


Figure 4. Top: the distractor amplitude of each session, in the order the sessions were run. Middle and bottom: each session's unimanual PSE in Model D, which assumes no trend (grey, 95% intervals), with the typical-session PSE of the two extensions: the bias follows the distractor amplitude (D+crit, solid) or drifts linearly over time (D+order, dashed). Both lines track the rise; only the four late 24 µm sessions could separate them.

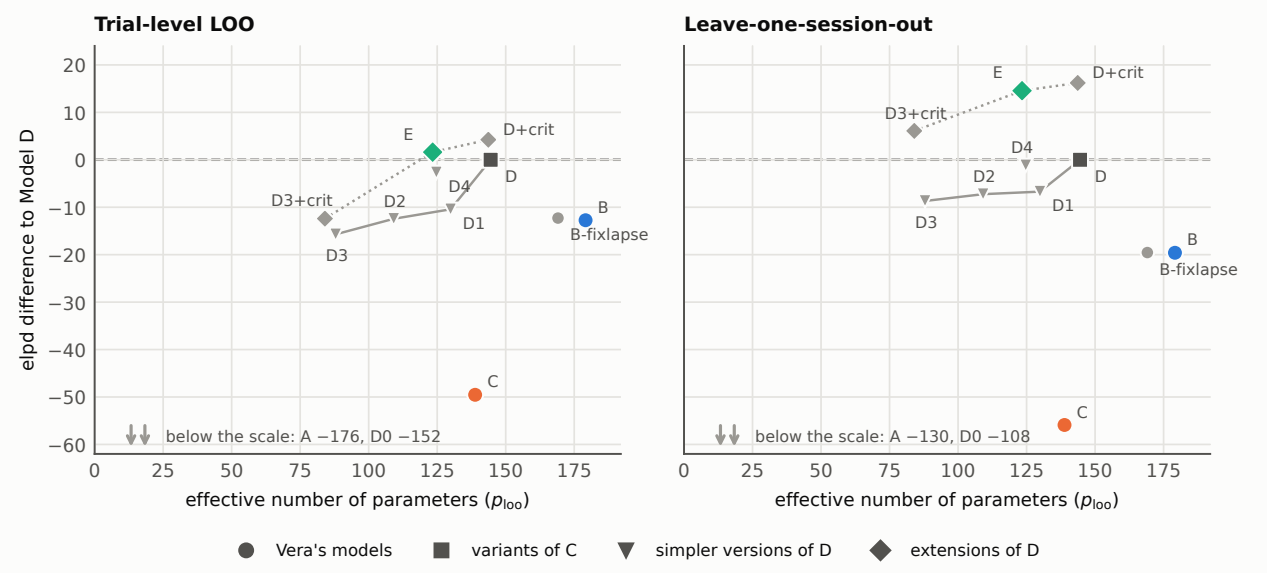


Figure 5. Predictive performance against model complexity (effective number of parameters, p_{loo}), relative to Model D (dashed line). Solid grey line: the simpler versions of D, from D3 (only the bias varies between sessions) through D2 and D1 to D. Dotted grey line: the same with the criterion trend, from D3+crit through E to D+crit.

2.4 Simpler models: only the bias and the lapses need to vary between sessions

Model D lets eight parameters per hand vary from session to session. The simpler versions hold some of them fixed (Fig. 5, Table 1).

- **The bias must vary.** With no session differences at all, D0 minus D is −152.0 at the trial level.
- **The distractor terms need not vary.** Holding b_2 and b_3 fixed for each hand (D4 minus D: -2.7 ± 3.7 at the trial level, -1.2 ± 2.8 for new sessions) saves 4 parameters per session and 20 effective parameters (145 → 125). Once its amplitude is taken into account, the distractor has the same effect in every session.
- **Neither, largely, does the slope.** Holding it fixed as well changes little (D3 minus D2: -3.3 ± 3.3 at the trial level, -1.4 ± 2.1 for new sessions).
- **The lapse rates do vary.** Holding them fixed loses ground (D1 minus D: -10.4 ± 5.7 at the trial level, -6.7 ± 3.6 for new sessions), and this step accounts for most of what the simpler models D2 and D3 lose.

The same simplifications hold with the criterion trend. **Model E** (D4+crit: bias, slope and lapses vary between sessions, the distractor terms are fixed per hand, and the bias follows the distractor amplitude) is as good as D+crit (E minus D+crit: -2.6 ± 4.0 at the trial level, -1.6 ± 2.9 for new sessions) with 20 fewer effective parameters (123 against 144). It beats D by 14.6 ± 5.6 and B by 34.2 ± 10.6 for new sessions. Even D3+crit, with a single bias per hand and session (2 parameters per session against 16 in B and D), predicts new sessions as well as D (6.1 ± 6.5) and better than B (25.7 ± 12.7), with fewer than half of B's effective parameters (84 against 179).

Table 2. Distractor effects in µm (posterior mean and 95% interval): the mean over all sessions, or over the sessions with a 6 or a 32 µm distractor. PSE shift: bimanual minus unimanual PSE; JND change: bimanual minus unimanual JND. In C the two curves differ only by a shift, so its JND change is zero by construction.

Quantity (µm)	Hand	B	C	D	E
PSE shift, mean	left	−4.3 [−5.1, −3.5]	−4.5 [−5.1, −3.8]	−4.0 [−4.8, −3.3]	−4.0 [−4.7, −3.3]
	right	−6.1 [−6.8, −5.3]	−5.7 [−6.4, −5.1]	−6.0 [−6.7, −5.2]	−5.9 [−6.7, −5.3]
PSE shift, 6 µm	left	−1.1 [−2.7, 0.5]	−1.3 [−1.8, −0.9]	−1.8 [−3.3, −0.5]	−1.4 [−2.7, −0.1]
	right	−3.9 [−5.3, −2.1]	−1.8 [−2.2, −1.4]	−4.2 [−5.6, −2.9]	−3.6 [−5.0, −2.3]
PSE shift, 32 µm	left	−6.6 [−8.0, −5.4]	−7.4 [−8.5, −6.3]	−5.7 [−6.8, −4.5]	−6.1 [−7.2, −5.0]
	right	−7.9 [−9.0, −6.7]	−8.6 [−9.7, −7.5]	−7.4 [−8.5, −6.3]	−7.7 [−8.7, −6.6]
JND change, mean	left	1.0 [0.5, 1.6]	0 (by design)	0.3 [0.2, 0.5]	0.4 [0.2, 0.5]
	right	0.5 [0.0, 1.1]	0 (by design)	0.2 [0.1, 0.3]	0.2 [0.1, 0.4]

2.5 The distractor effects in the recommended model

Every model that predicts well gives the same picture of the distractor's effect on the point of subjective equality (PSE, the target amplitude judged "high" half the time; Table 2, Fig. 6).

- On bimanual trials the PSE moves down. Averaged over sessions, the shift in Model E is −4.0 µm for the left hand (95% interval −4.7 to −3.3) and −5.9 µm for the right (−6.7 to −5.3). A distractor makes the target feel stronger.
- The shift grows with distractor amplitude, from −1.4 µm in the 6 µm sessions to −6.1 µm in the 32 µm sessions for the left hand, and from −3.6 to −7.7 µm for the right.
- The right hand is shifted even by a weak distractor: its presence offset is $b_3 = 0.76$ (0.24 to 1.29), against 0.15 (−0.40 to 0.66) for the left.
- There is no reliable change in sensitivity. Model B's increase of the left hand's JND, 1.03 µm (0.48 to 1.59), shrinks to 0.35 µm in E, and that remainder comes from the bimanual lapses, which flatten the curve's ends, not from a change of slope. With a free bimanual slope the increase is 0.78 µm, but the slope change itself is uncertain ($\mu_\delta = -0.40$, −1.08 to 0.30) and does not improve prediction.

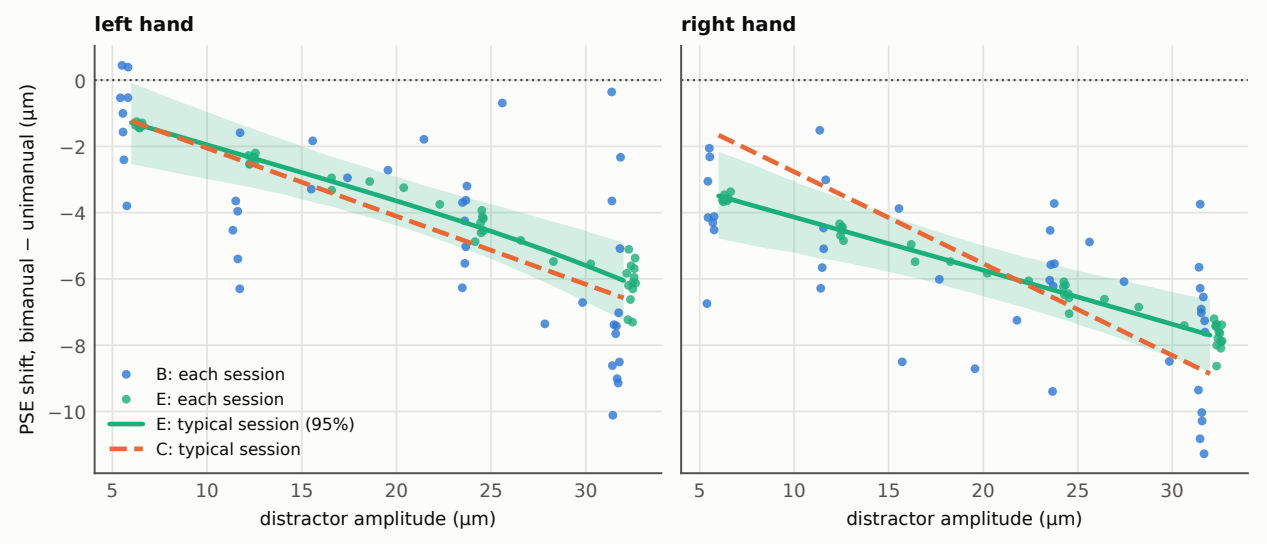


Figure 6. PSE shift (bimanual minus unimanual) against the session's distractor amplitude. Dots: each session's estimate in Models B and E. Lines: a typical session in E (with 95% interval) and in C. C forces the shift to be proportional to amplitude; E allows a presence offset, which the right hand needs.

3 What the models say about interference

A distractor makes the target feel stronger, and more so the stronger it is. Every model that predicts well gives the same average bimanual shift, about 4 µm for the left hand and 6 µm for the right, and the shift grows with the distractor amplitude. This is what one expects if the vibrations on the two hands are partly summed before the judgement. For the left hand the shift is proportional to the amplitude. The right hand has, in addition, a shift that does not depend on amplitude; this could be a real difference between the hands or a shift that rises quickly at low amplitudes and then grows more slowly. With one monkey the two cannot be told apart. A shift of the form $b_2\, d^{\,p}$ would be the natural next variant to test.

Sensitivity does not change. The slope of the psychometric curve is the same with and without the distractor. The left-hand JND increase in Model B comes from the bimanual lapses, which B can only absorb into a flatter curve. The claim that the distractor reduces sensitivity should be dropped, or stated as not detected.

The extra errors look like answers driven by the distractor. The bimanual "high" errors to weak targets are not spread evenly. For the left hand they occur only when the distractor is strong (24–32 µm, near or above the 28 µm boundary), which suggests that on a few percent of trials a strong distractor decides the answer. Their response times point the same way. In the 24–32 µm sessions these errors are only 35–50 ms slower than correct answers (left hand 615 against 564 ms, right 598 against 562 ms). Errors that cannot come from the distractor are much slower: "low" answers to strong targets take about 805 ms against 546 ms, and the right hand's weak-target errors in the 6–12 µm sessions about 1,010 against 550 ms. A literal "judges the distractor" model does not fit, though. Sirius calls a 24–32 µm vibration "high" about half of the time or less (its unimanual PSE is near 32 µm), so judging such a distractor would add "low" answers to strong targets about as often as "high" answers to weak ones; strong targets drew few "low" answers in those sessions (Fig. 2, bottom).

The errors are better described as occasional "high" answers pushed by a strong distractor than as reports of the distractor itself, and neither capture model fits better than D's amplitude-dependent lapse rates. The choices alone do not settle the mechanism; the response times are the more direct test (Section 4).

The criterion moved between sessions. Over the four months Sirius became more reluctant to call a unimanual target "high": its PSE rose by about 3.0 µm for the left hand and 6.1 µm for the right (amplitude version; slightly more in the time version). Two readings fit the data equally well. Sirius may have adjusted its criterion to the distractor: in sessions with strong distractors many bimanual targets feel stronger, and a monkey that keeps its overall rate of "high" answers steady would raise its criterion, as the left hand's numbers suggest. Or the criterion simply drifted, for reasons unrelated to the distractor. The design cannot separate the two, because the distractor amplitude rose with time. Either way, the trend matters for the distractor effects: without it, part of the trend is mistaken for a presence effect.

Model choice. Model B spends its parameters on things the data do not need, such as separate slopes and biases for unimanual and bimanual trials in every session, and misses what they do need: lapse rates tied to the distractor and a trend in the criterion. Model E keeps what is needed (a bias, slope and lapse rates for each session, bimanual lapses that depend on the distractor, a presence offset and the criterion trend) and fixes the rest.

4 Further extensions worth trying

These were not fitted here. Each stays close to the current models and uses data that already exist.

1. **Use the frequency-probe trials.** Vera's selection drops 3,232 completed trials in which a single vibration was given at 150 or 250 Hz instead of 200 Hz (all unimanual, from the same days). With a frequency term in η they would add about 75 unimanual trials per session, which would sharpen each session's bias, slope and unimanual lapses: the baseline against which every bimanual effect is measured. They also test whether Sirius judges amplitude or perceived intensity, which bears on whether the distractor adds to the target's intensity.
2. **Response times.** The trial files record when the choice targets appear and when the response movement starts (median response time 562 ms). A first look supports treating some bimanual "lapses" as real decisions driven by the distractor (Section 3). A drift-diffusion version of Model E, in which the distractor adds to the drift rate and lapses are a separate response process, would use both choices and times and could tell a change in the evidence from a change in the criterion.
3. **A more flexible trend in the criterion.** The linear trend and the amplitude term fit equally well. A random walk over sessions would let the criterion wander without assuming a shape, and a criterion that follows a running average of recent distractor amplitudes would describe slow adaptation. On these data they would be hard to tell apart for the same reason as D+crit and D+order, but they are the right models for data in which amplitude and time are not confounded.
4. **Trial history.** A logistic check with the full session-by-hand structure finds only a weak tendency to switch away from the previous choice (logit -0.20 ± 0.11) and no clear win-stay

effect (0.13 ± 0.12). Adding the previous choice to η is cheap, but it is unlikely to change the conclusions.

5. **Correlated session effects.** The models draw each session parameter independently. A full covariance across a session's parameters (an LKJ prior on bias, slope and lapses, and across the two hands) is the general form of the shared daily bias tested in D+shared.

6. **More animals.** A monkey level above the session level would separate effects that hold for every animal from the individual pattern seen here, such as the right hand's extra shift.

5 Recommendations

1. **Use Model E as the main model** (D with the distractor terms fixed per hand and the criterion trend). It predicts as well as the best model found and is simpler than D. Report both scores: trial-level LOO with session-clustered standard errors, and leave-one-session-out.

2. **Report from E**: the average PSE shift and its growth with amplitude; the right hand's extra shift for any distractor; the bimanual "high" errors that depend on hand and amplitude; and the criterion trend. Drop the claim of reduced sensitivity, or state it as not detected.

3. **Do not attribute the criterion trend to the distractor** without new data. If more sessions are run, vary the distractor amplitude across sessions in random order, or interleave several amplitudes within a session, so that adaptation can be separated from drift.

4. **Before the slides are updated**, run a convergence check of E at its fitted values and a model-recovery study of E against B, as Vera did for B against A.

5. **Next extensions**: the frequency-probe trials and the response times (Section 4).

A Model definitions

Common parts. Every model gives the probability that the monkey answers "high" on trial i as

$$P(y_i = \text{high}) \;=\; \gamma_h + \big(1 - \gamma_h - \gamma_l\big)\ \text{logistic}(\eta_i), \qquad \gamma = 0.25\ \text{logistic}(\ell),$$

where γ_h and γ_l are the "high" and "low" lapse rates (each at most 0.25). The target amplitude a_i enters as $x_i = (a_i - 28)/14.63$ (14.63 µm is the standard deviation of the eight amplitudes), the distractor as $d_i = a_i^{\text{dist}}/14.63$ on bimanual trials and $d_i = 0$ on unimanual trials, and $c_i = 1$ marks bimanual trials. h is the stimulated hand, s the session, d_s the scaled distractor amplitude of session s and $\bar{d}$ its mean over the 42 sessions. Session-level parameters are drawn from normal distributions with hand-level (or group-level) means μ and standard deviations σ, written non-centred ($b = \mu + \sigma z$, $z \sim N(0,1)$).

Vera's models. **A**: four groups g (hand × condition), $\eta = b_{0g} + b_{1g}x$, lapses per group, no session differences; $b_0 \sim N(0, 6^2)$, $b_1 \sim N(4, 2^2)$, $\gamma/0.25 \sim$ Beta with mean 0.2 and sd 0.15. **B**: the same four groups, each with its own hierarchy over sessions: $\eta = b_{0gs} + b_{1gs}x$, lapses ℓ_{gs}. **C**: one hierarchy per hand: $\eta = b_{0hs} + b_{1hs}x + b_{2hs}d$, with lapses ℓ_{hs} shared by unimanual and bimanual trials. B and C use the same hyperpriors: $\mu_{b_0} \sim N(0, 2^2)$, $\sigma_{b_0} \sim \text{HalfNormal}(2)$, $\mu_{b_1} \sim \text{LogNormal}(\log 4, 0.5^2)$, $\sigma_{b_1} \sim \text{Exp}(1)$, $\mu_\ell \sim N(-3, 1.5^2)$, $\sigma_\ell \sim \text{HalfNormal}(0.5)$; in C also $\mu_{b_2} \sim N(0,1)$, $\sigma_{b_2} \sim \text{HalfNormal}(1)$.

B-wide widens B's lapse prior to $\sigma_\ell \sim \text{HalfNormal}(1.5)$; **B-fixlapse** gives each group one lapse pair for all sessions.

Additions to C (one letter per addition). **L**: bimanual trials get their own lapse rates, $\ell^{\text{bi}}_{hs} = \mu^{\text{bi}}_h + \sigma^{\text{bi}}_h z$, with $\mu^{\text{bi}} \sim N(-3, 1.5^2)$ and $\sigma^{\text{bi}} \sim \text{HalfNormal}(0.5)$. **K**: these bimanual lapse means change with the session's distractor amplitude, $\mu^{\text{bi}}_h + \kappa_h(d_s - \bar{d})$, $\kappa \sim N(0, 2^2)$, separately for "high" and "low" lapses. **P**: a presence offset, $\eta \mathrel{+}= b_{3hs}c$, with $\mu_{b_3} \sim N(0,1)$, $\sigma_{b_3} \sim \text{HalfNormal}(1)$; the bimanual shift is then $b_3 + b_2 d$ instead of being proportional to d. **S**: a bimanual slope change, $\eta \mathrel{+}= \delta_{hs}\, c\, x$, $\mu_\delta \sim N(0,1)$, $\sigma_\delta \sim \text{HalfNormal}(1)$. **M** (capture): on a fraction π_h of bimanual trials the monkey answers from its unimanual curve evaluated at the stronger of the two vibrations (**Mmax**) or at the distractor (**Mdist**); $\text{logit}\,\pi \sim N(-2.5, 1.5^2)$.

Model D and its simpler versions. **D** = C+LKP: $\eta = b_{0hs} + b_{1hs}x + b_{2hs}d + b_{3hs}c$, unimanual lapses ℓ_{hs} as in C, and bimanual lapses $\ell^{\text{bi}}_{hs} = \mu^{\text{bi}}_h + \kappa_h(d_s - \bar{d}) + \sigma^{\text{bi}}_h z$. The simpler versions set some session standard deviations to zero, so that parameter takes one value per hand: **D4** the distractor terms b_2, b_3; **D1** the lapses (both conditions; κ kept); **D2** both of these; **D3** as D2 and also the slope b_1, so only the bias b_0 differs between sessions; **D0** everything (no session differences).

Extensions of D. **D+crit**: the hand-level mean of the session bias depends on the session's distractor amplitude, $\mu_{b_0,h} + \lambda_h(d_s - \bar{d})$, $\lambda \sim N(0,1)$. Because b_0 enters unimanual and bimanual trials alike, this moves the whole session's criterion, not the bimanual shift. **D+order**: instead a linear trend over the study, $\mu_{b_0,h} + \tau_h(s - 20.5)/10$ with $s = 0, \ldots, 41$ the session's position, $\tau \sim N(0,1)$. **D+crit+order**: both terms. **D+shared**: a daily offset shared by the two hands, $b_{0hs} + v_s$, $v_s \sim N(0, \sigma_v^2)$, $\sigma_v \sim \text{HalfNormal}(1)$. The same terms were added to the simpler versions: **E** = D4+crit, **D3+crit**, and **E+shared** (E with the shared offset). All other priors are as in C.

B Sampling diagnostics and checks of the comparison

Table 3 lists, for every model, the sampler, the run time on two cores, the number of divergent transitions, the largest $\hat{R}$, the smallest bulk and tail effective sample sizes over all parameters (Model B's sign-free z_σ variables excluded, as in Vera's reports), and the largest Pareto k of the two cross-validation scores. Trial-level LOO has every k below 0.7 for every model.

Checks of the leave-one-session-out scores. The LOSO score of a session is an importance-sampling estimate, so it was checked directly in the earlier 1000-draw runs: for four held-out sessions the model was refit on the other 41 sessions and the session predicted from the refit. The exact and estimated scores agreed to within 0.2 (Model B, session 05-12: -99.64 vs -99.59; Model D, 05-12: -101.54 vs -101.41; C+L, 04-10: -61.26 vs -61.29 with $k = 1.03$; C+LKPS, 04-10: -60.93 vs -60.76 with $k = 0.79$). A second random seed changed the totals by less than 0.1 in those runs, and by at most 0.01 for B, D, D+crit and E in the 5000-draw runs. Sessions with $k > 0.7$ in Table 3 are therefore not a concern at the precision used in this report.

1000 versus 5000 draws. For the 14 models fit both ways, the trial-level elpd changed by at most 1.04 (median 0.37) and the LOSO elpd by at most 0.67 (median 0.30). No pair of models changed order under trial-level LOO, and one pair changed order under LOSO (C+L and C+LMmax, 0.4 apart before and 0.4 after).

Table 3. Sampling diagnostics (4 chains × 5000 draws after 1000 tuning steps).

Model	Sampler	Minutes	Div.	max $\hat{R}$	min ESS bulk	min ESS tail	max k trial	max k LOSO	sessions $k > 0.7$
A	PyMC	4	0	1.0005	14957	8194	0.16	0.44	0
B	PyMC	12	0	1.0010	3496	3101	0.49	0.85	1
B-wide	PyMC	20	0	1.0017	2849	4407	0.63	0.60	0
B-fixlapse	PyMC	12	0	1.0018	5076	8517	0.51	0.67	0
C	nutpie	8	0	1.0015	2385	3027	0.35	0.67	0
C+L	nutpie	9	0	1.0015	3365	3391	0.44	0.69	0
C+LK	nutpie	8	0	1.0015	3139	4065	0.47	0.83	2
C+LP	nutpie	14	0	1.0018	2738	2099	0.50	0.81	1
C+LPS	nutpie	10	0	1.0032	2315	2282	0.44	0.55	0
C+LKPS	nutpie	10	0	1.0017	2537	1913	0.39	0.55	0
C+Mmax	nutpie	10	0	1.0012	2318	2589	0.41	0.58	0
C+Mdist	nutpie	9	0	1.0035	2439	2884	0.40	0.83	1
C+LMmax	nutpie	11	0	1.0020	2892	3017	0.43	0.79	1
C+LPMmax	nutpie	12	0	1.0021	2896	3123	0.43	0.61	0
C+LPMdist	nutpie	12	0	1.0021	2754	3079	0.44	0.55	0
D (= C+LKP)	nutpie	13	0	1.0015	3027	3217	0.44	0.71	1
D4	nutpie	7	0	1.0022	4003	4821	0.46	0.97	2
D1	nutpie	9	0	1.0017	3030	3841	0.40	0.57	0
D2	nutpie	6	0	1.0032	2377	2087	0.38	0.67	0
D3	nutpie	5	0	1.0009	2674	4703	0.38	0.49	0
D0	nutpie	5	0	1.0007	8077	6401	0.31	1.07	1
D+shared	nutpie	10	0	1.0015	2172	1877	0.43	0.68	0
D+crit	nutpie	9	0	1.0027	2878	2725	0.44	0.59	0
D+order	nutpie	8	0	1.0014	2797	2828	0.44	1.03	1
D+crit+order	nutpie	10	0	1.0019	2722	2533	0.43	0.68	0
E (= D4+crit)	nutpie	9	0	1.0013	3570	3720	0.43	0.58	0
D3+crit	nutpie	5	0	1.0014	4235	6151	0.48	0.78	1
E+shared	nutpie	8	0	1.0041	1735	1439	0.46	0.62	0